

Id no:25dce049

Practical 3: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition Design responsive layouts for Home, About, Registration, Dashboard, and

Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

Key Questions / Analysis

/ Interpretation

- 1. How does the layout adapt for mobile, tablet, and desktop screens?**
- 2. Which layout technique is used and why?**
- 3. Are typography, spacing, colors, and contrast readable?**
- 4. Is the framework used consistently without unnecessary inline CSS?**

Key Skills to be

Addressed

CSS3, media queries, Flexbox, Grid, Bootstrap components, responsive design

Applications Multi-device web interface design

Learning Outcome Students will design a professional, responsive, and reusable UI layout.

CO Mapping CO2

Dataset/Test Data N/A; UI design only

Tools/Technology to be

Used

HTML5, CSS3, Bootstrap, browser responsive mode

Total Hours Implementation: 4 hours | Total engagement: 6 hours

Post Laboratory Work

Description

Submit screenshots for mobile, tablet, and desktop views.

Evaluation Strategy

Including Viva

Responsiveness, framework usage, visual consistency, and CSS organization.

Advanced/Intermediate

Extension

Intermediate: Add Contact and Feedback pages.

Advanced: Create reusable layout components using JavaScript libraries such as jQuery.

Code:

Style.css

```
* {  
    box-sizing: border-box;  
    margin: 0;  
    padding: 0;  
}
```

```
body {  
    font-family: Arial, sans-serif;  
    line-height: 1.6;  
    background-color: #f4f6f8;  
    color: #222;  
}
```

```
header {  
    background-color: #1e3a8a;  
    color: white;  
    padding: 20px;  
}
```

```
header h1 {
```

```
    margin-bottom: 15px;
}
```

```
nav {
    display: flex;
    flex-direction: column;
    gap: 10px;
}
```

```
nav a {
    color: white;
    text-decoration: none;
}
```

```
main {
    width: 90%;
    max-width: 1200px;
    margin: auto;
    padding: 30px 0;
}
```

```
.hero {  
    text-align: center;  
    padding: 40px 10px;  
}
```

```
.cards {  
    display: grid;  
    grid-template-columns: 1fr;  
    gap: 20px;  
    margin-top: 25px;  
}
```

```
.card {  
    background-color: white;  
    padding: 20px;  
    border-radius: 8px;  
}
```

```
button {
```

```
padding: 10px 20px;
border: none;
border-radius: 5px;
cursor: pointer;
}
```

```
footer {
    text-align: center;
    padding: 20px;
    background-color: #1e3a8a;
    color: white;
}
```

```
/* Tablet */
```

```
@media (min-width: 768px) {

    nav {
        flex-direction: row;
    }
}
```

```
.cards {  
  grid-template-columns: repeat(2, 1fr);  
}  
}
```

```
/* Desktop */
```

```
@media (min-width: 1024px) {
```

```
  header {  
    display: flex;  
    justify-content: space-between;  
    align-items: center;  
  }
```

```
  header h1 {  
    margin-bottom: 0;  
  }
```

```
.cards {  
  grid-template-columns: repeat(3, 1fr);
```

```
}  
}
```

index.html:

```
<!doctype html>  
<html lang="en">  
  <head>  
    <meta charset="UTF-8" />  
    <meta name="viewport" content="width=device-width, initial-scale=1.0"  
  />  
    <title>StudentHub - Contact</title>  
  
    <!-- <link rel="stylesheet" href="/css/style.css" /> -->  
    <link rel="stylesheet" href="pages/css/style.css" />  
  
    <link  
      rel="stylesheet"  
  
      href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.6.0/css/all.min.  
      css"  
    />  
  </head>  
  
  <body>  
    <div class="sidebar">  
      <h1 class="logo">StudentHub</h1>
```

```
<ul>
  <li>
    <a href="index.html"><i class="fa fa-home"></i> Home</a>
  </li>
  <li>
    <a href="profile.html"><i class="fa fa-user"></i> Profile</a>
  </li>
  <li>
    <a href="courses.html"><i class="fa fa-book"></i> Course</a>
  </li>
  <li>
    <a href="assignments.html"><i class="fa fa-file"></i>
Assignment</a>
  </li>
  <li>
    <a href="attendance.html"
    ><i class="fa fa-calendar-check"></i> Attendance</a>
  </li>
  <li>
    <a href="result.html"><i class="fa fa-chart-bar"></i> Results</a>
  </li>
  <li class="active">
    <a href="contact.html"><i class="fa fa-phone"></i> Contact</a>
```

```
</li>
</ul>
</div>
```

```
<div class="main">
  <div class="topbar">
    <div class="search">
      <i class="fa fa-search"></i>
      <input type="text" placeholder="Search" />
    </div>
```

```
    <div class="profile-icon">
      <i class="fa-solid fa-user"></i>
    </div>
  </div>
```

```
<div class="container">
  <h2>Contact Us</h2>
```

```
<p>
  If you have any questions, suggestions, or issues, feel free to
  contact us.
</p>
```

```
<div class="contact-box">
```

```
<div class="info">
```

```
<h3>Contact Information</h3>
```

```
<p><strong>Email:</strong> support@studenthub.com</p>
```

```
<p><strong>Phone:</strong> +91 98765 43210</p>
```

```
<p>
```

```
<strong>Address:</strong> CHARUSAT University, Changa,  
Anand,
```

```
Gujarat
```

```
</p>
```

```
</div>
```

```
<div class="form">
```

```
<h3>Send Message</h3>
```

```
<form>
```

```
<input type="text" placeholder="Full Name" required />
```

```
<input type="email" placeholder="Email Address" required />
```

```
<input type="text" placeholder="Subject" required />
```

```

<textarea rows="6" placeholder="Your Message"></textarea>

<button type="submit">Send Message</button>

</form>

</div>

</div>

</div>

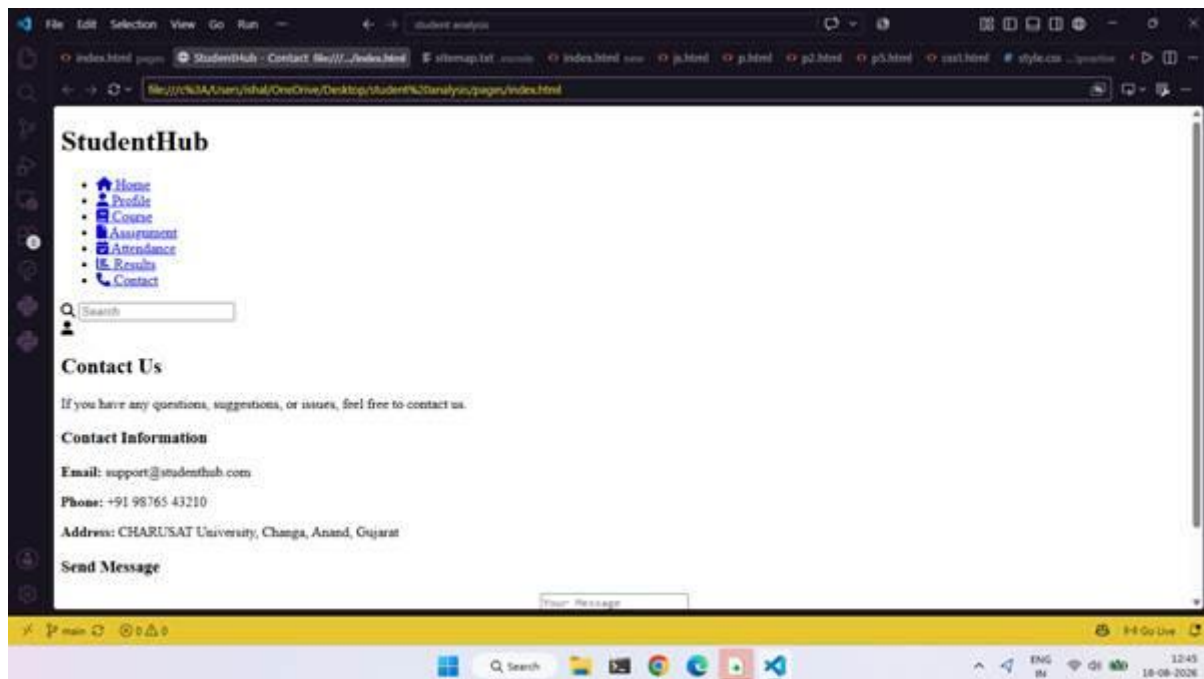
</div>

</body>

</html>

```

Output:



Login.html:

```
<!doctype html>
```

```
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
  />
    <title>StudentHub | Login</title>
    <link rel="stylesheet" href="css/style.css" />
  </head>

  <body>

    <header>
      <h1>StudentHub</h1>
      <p>Student Portal Management System</p>
    </header>
  </body>

  <!-- Navigation -->
  <nav>
    <a href="index.html">Home</a>
    <a href="about.html">About</a>
    <a href="register.html">Register</a>
    <a href="login.html">Login</a>
    <a href="dashboard.html">Dashboard</a>
    <a href="events.html">Events</a>
    <a href="profile.html">Profile</a>
```

```
<a href="contact.html">Contact</a>
<a href="faq.html">FAQ</a>
<a href="feedback.html">Feedback</a>
</nav>

<!-- Main Content -->
<main>
  <section>
    <h2>Student Login</h2>

    <form action="#" method="post">
      <label for="email">Email Address</label><br />
      <input
        type="email"
        id="email"
        name="email"
        placeholder="Enter your email"
        required
      />
      <br /><br />

      <label for="password">Password</label><br />
      <input
        type="password"
```

```
id="password"
name="password"
placeholder="Enter your password"
minlength="6"
required
/>
<br /><br />

<label>
  <input type="checkbox" name="remember" />
  Remember Me
</label>

<br /><br />

<input type="submit" value="Login" />
<input type="reset" value="Clear" />
</form>

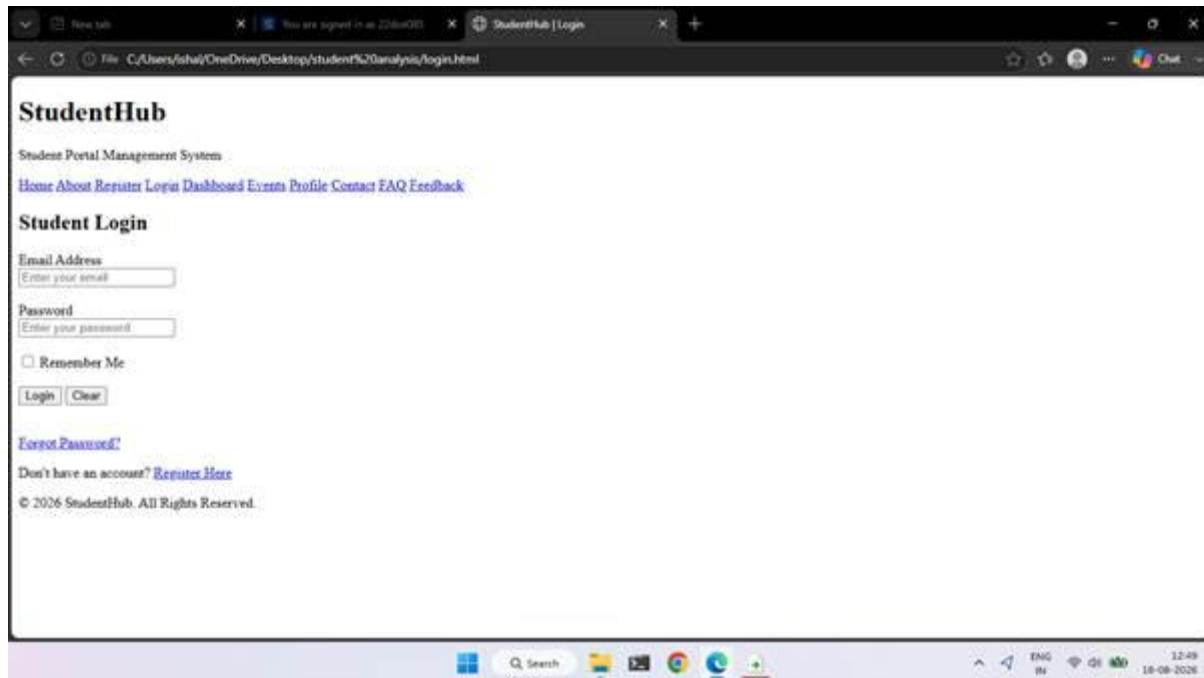
<br />

<p>
  <a href="#">Forgot Password?</a>
</p>
```

```
<p>
  Don't have an account?
  <a href="register.html">Register Here</a>
</p>
</section>
</main>

<!-- Footer -->
<footer>
  <p>&copy; 2026 StudentHub. All Rights Reserved.</p>
</footer>
</body>
</html>
```

Output:



Courses.html:

```
<!doctype html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>StudentHub - Courses</title>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>StudentHub</h1>
```

```
<nav>
  <a href="index.html">Home</a> | <a href="about.html">About</a> |
  <a href="register.html">Register</a> | <a href="login.html">Login</a>
|
  <a href="dashboard.html">Dashboard</a> |
  <a href="events.html">Events</a> | <a
href="courses.html">Courses</a> |
  <a href="assignments.html">Assignments</a> |
  <a href="attendance.html">Attendance</a> |
  <a href="profile.html">Profile</a> |
  <a href="contact.html">Contact</a> |
  <a href="faq.html">FAQ</a>
</nav>
```

```
<hr />
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<h2>Available Courses</h2>
```

```
<table border="1" cellpadding="10">
```

```
<thead>
```

```
<tr>
```

```
<th>Course Code</th>
```

```
<th>Course Name</th>
<th>Faculty</th>
<th>Credits</th>
</tr>
</thead>

<tbody>
<tr>
<td>CSE101</td>
<td>Programming in C</td>
<td>Dr. Mehta</td>
<td>4</td>
</tr>

<tr>
<td>CSE202</td>
<td>Object Oriented Programming with Java</td>
<td>Prof. Patel</td>
<td>4</td>
</tr>

<tr>
<td>CSE203</td>
<td>Web Development</td>
```

```
<td>Prof. Shah</td>
<td>3</td>
</tr>

<tr>
<td>CSE204</td>
<td>Database Management System</td>
<td>Dr. Trivedi</td>
<td>4</td>
</tr>

<tr>
<td>CSE205</td>
<td>Data Structures</td>
<td>Prof. Desai</td>
<td>4</td>
</tr>
</tbody>
</table>
</section>
</main>

<footer>
<hr />
```

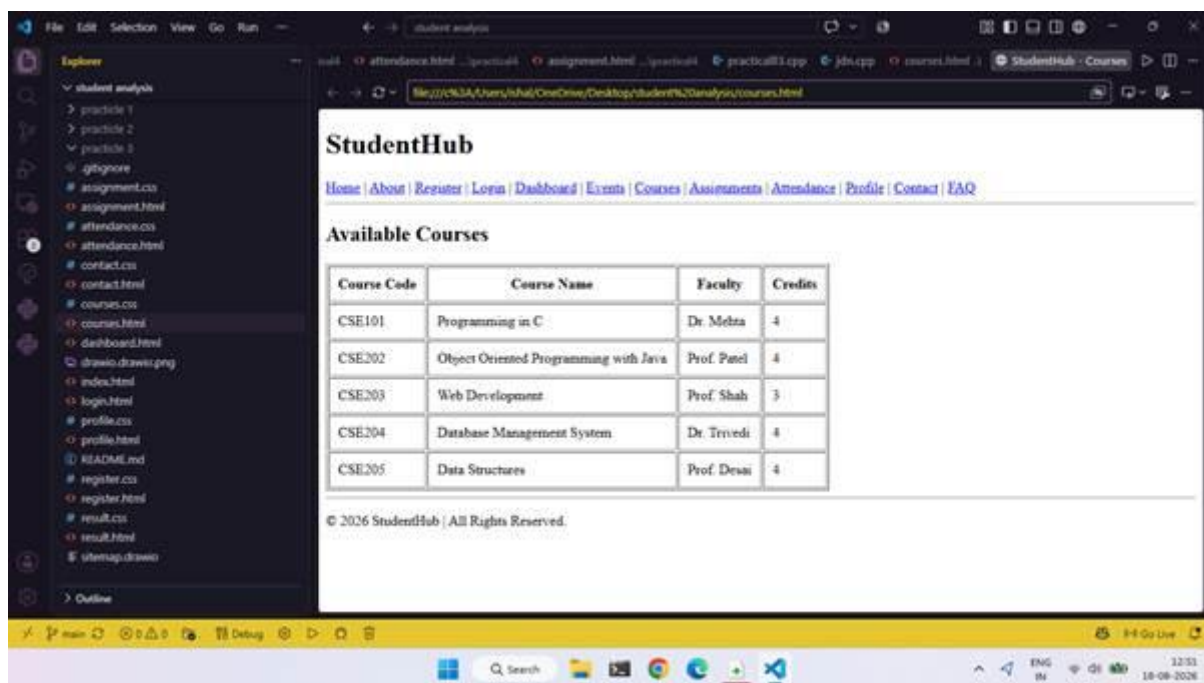
<p>© 2026 StudentHub | All Rights Reserved.</p>

</footer>

</body>

</html>

Output:



Attendance.html

<!doctype html>

<html lang="en">

<head>

```
<meta charset="UTF-8" />
<meta name="viewport" content="width=device-width, initial-scale=1.0"
/>
<title>StudentHub - Assignments</title>
</head>

<body>
  <header>
    <h1>StudentHub</h1>

    <nav>
      <a href="index.html">Home</a> | <a href="about.html">About</a> |
      <a href="register.html">Register</a> | <a href="login.html">Login</a>
|
      <a href="dashboard.html">Dashboard</a> |
      <a href="events.html">Events</a> | <a href="profile.html">Profile</a>
|
      <a href="assignments.html">Assignments</a> |
      <a href="attendance.html">Attendance</a> |
      <a href="contact.html">Contact</a> | <a
href="admin.html">Admin</a> |
      <a href="faq.html">FAQ</a> |
      <a href="feedback.html">Feedback</a>
    </nav>

    <hr />
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<h2>Assignment List</h2>
```

```
<table border="1" cellpadding="10">
```

```
<thead>
```

```
<tr>
```

```
<th>Subject</th>
```

```
<th>Assignment</th>
```

```
<th>Due Date</th>
```

```
<th>Status</th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr>
```

```
<td>Web Development</td>
```

```
<td>Semantic HTML Project</td>
```

```
<td>25 July 2026</td>
```

```
<td>Pending</td>
```

```
</tr>
```

<tr>

<td>Java Programming</td>

<td>OOP Concepts</td>

<td>28 July 2026</td>

<td>Submitted</td>

</tr>

<tr>

<td>Database Management</td>

<td>ER Diagram</td>

<td>30 July 2026</td>

<td>Pending</td>

</tr>

</tbody>

</table>

</section>

</main>

<footer>

<hr />

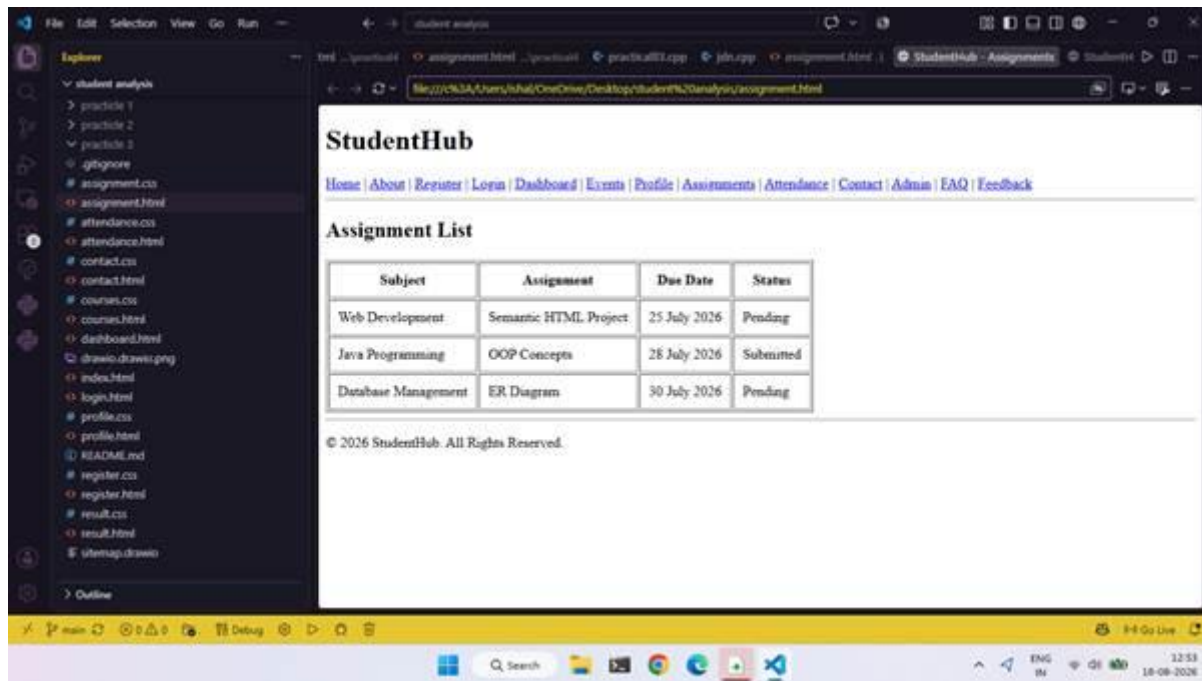
<p>© 2026 StudentHub. All Rights Reserved.</p>

</footer>

</body>

</html>

Output:



Profile.html

<header>

<table width="100%">

<tr>

<td width="80">

<img

alt="https://png.pngtree.com/png-vector/20190223/ourmid/pngtree-student-glyph-black-icon-png-image_691145.jpg"

width="70"

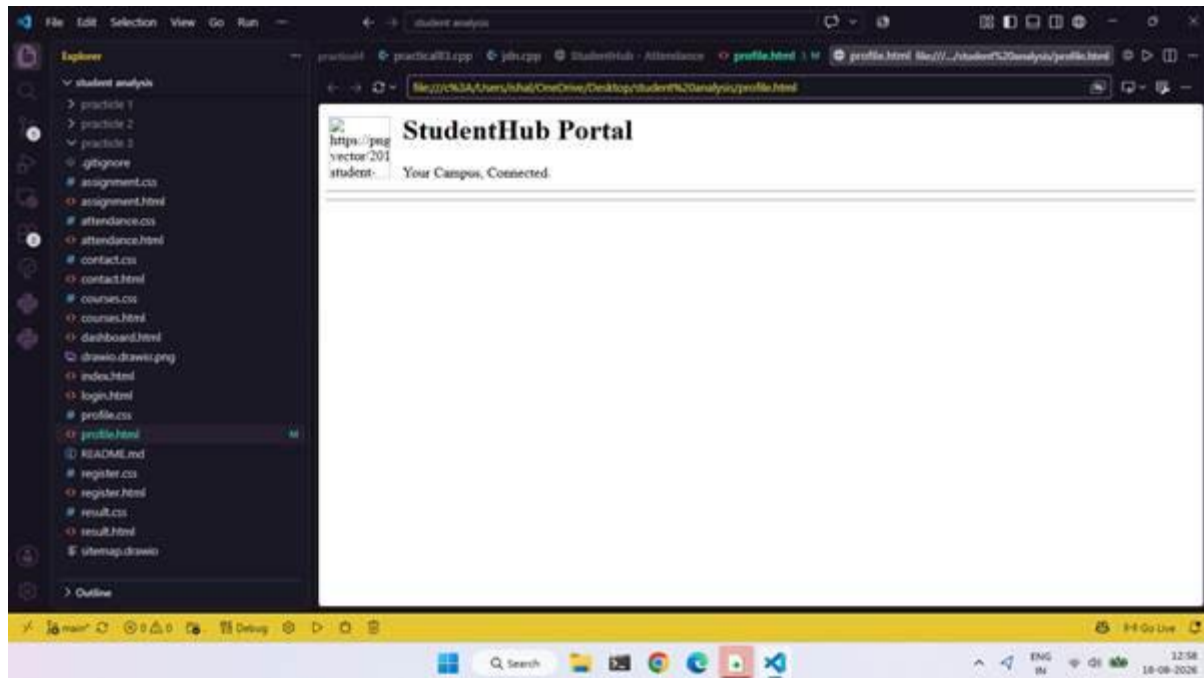
```
        height="70"
    />
</td>

<td>
    <h1>StudentHub Portal</h1>
    <p>Your Campus, Connected.</p>
</td>
</tr>
</table>

<hr />

<hr />
</header>
```

Output:



Register.html

```
<!doctype html>
```

```
<html lang="en">
```

```
<h2>Student Registration</h2>
```

```
<form>
```

```
<label for="name">Name</label><br />
```

```
<input type="text" id="name" /><br /><br />
```

```
<label for="email">Email</label><br />
```

```
<input type="email" id="email" /><br /><br />
```

```
<label for="password">Password</label><br />
```

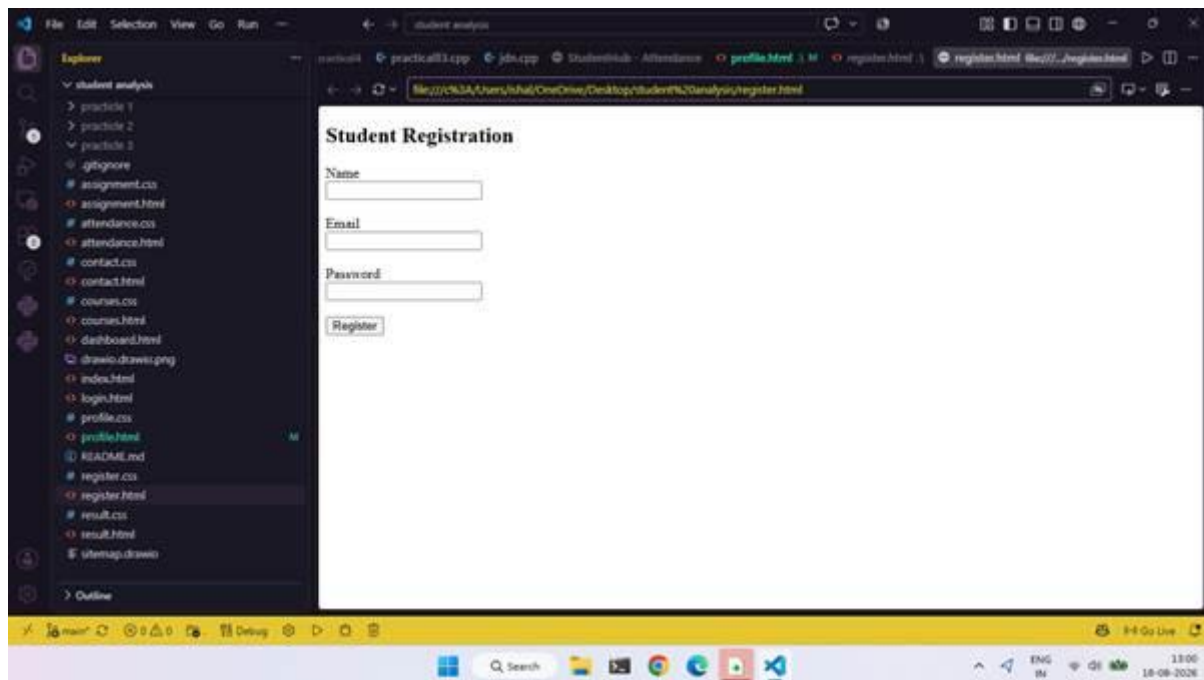
```
<input type="password" id="password" /><br /><br />
```

```
<button>Register</button>
```

```
</form>
```

```
</html>
```

Output:



Results.html

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<title>Feedback</title>
```

```
</head>
```

```
<body>
```

```
  <header>
```

```
    <h1>Feedback</h1>
```

```
    <nav></nav>
```

```
  </header>
```

```
  <main>
```

```
    <form>
```

```
      <label>Your Name</label><br />
```

```
      <input type="text" /><br /><br />
```

```
      <label>Comments</label><br />
```

```
      <textarea rows="5" cols="30"></textarea><br /><br />
```

```
      <button>Submit</button>
```

```
    </form>
```

```
  </main>
```

```
  <footer>
```

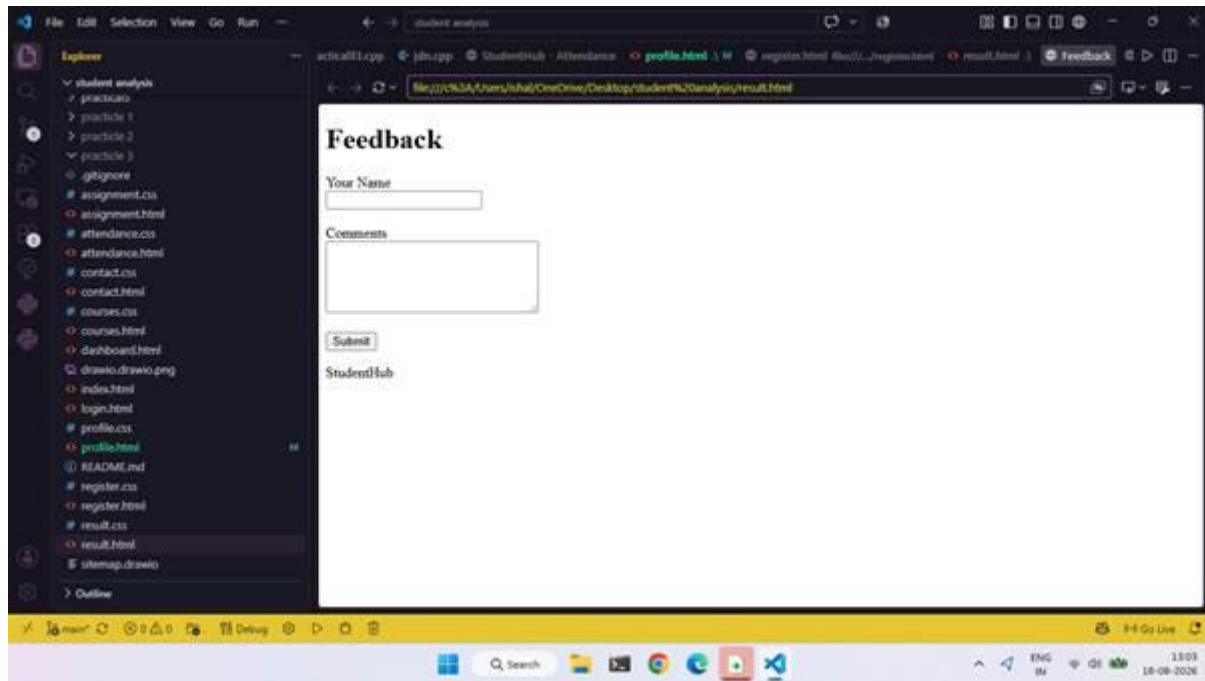
```
    <p>StudentHub</p>
```

</footer>

</body>

</html>

Output:



Dashboard.html

<!doctype html>

<html lang="en">

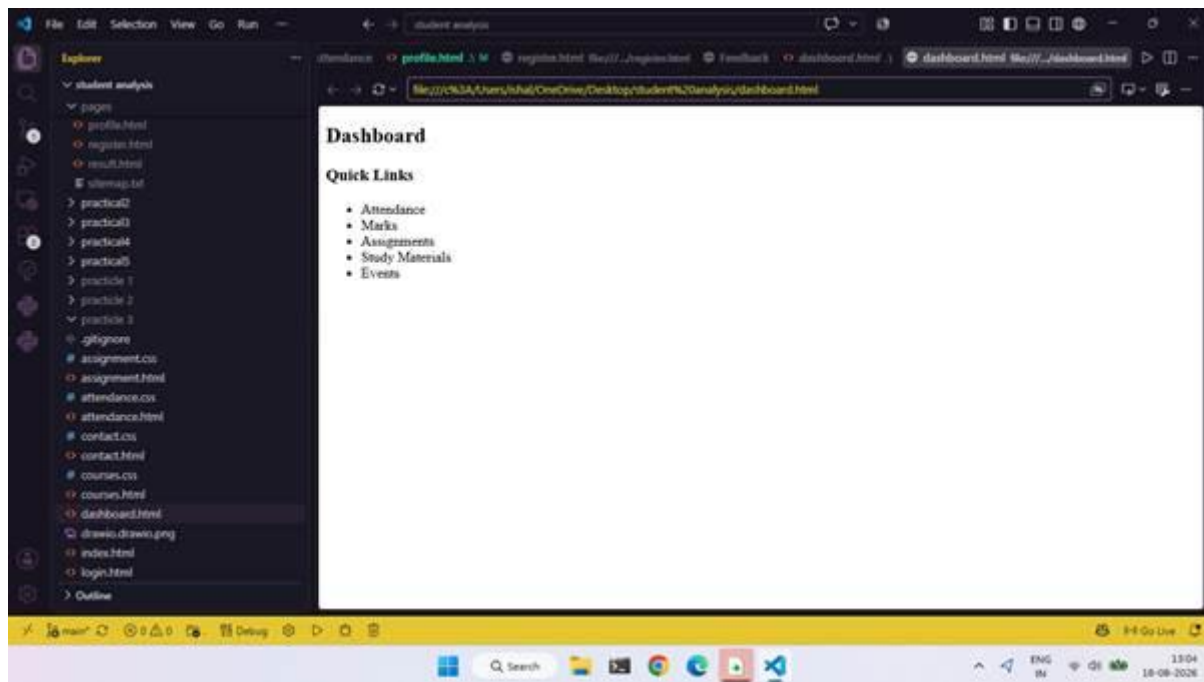
<h2>Dashboard</h2>

<section>

<h3>Quick Links</h3>

```
<ul>
  <li>Attendance</li>
  <li>Marks</li>
  <li>Assignments</li>
  <li>Study Materials</li>
  <li>Events</li>
</ul>
</section>
</html>
```

Output:



Contact.html

```
<!doctype html>
```

```
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0"
  />
    <title>StudentHub - Contact</title>
  </head>

  <body>
    <header>
      <h1>StudentHub</h1>

      <nav>
        <a href="index.html">Home</a> | <a href="about.html">About</a> |
        <a href="register.html">Register</a> | <a href="login.html">Login</a>
      |
        <a href="dashboard.html">Dashboard</a> |
        <a href="events.html">Events</a> | <a
href="courses.html">Courses</a> |
        <a href="assignments.html">Assignments</a> |
        <a href="attendance.html">Attendance</a> |
        <a href="profile.html">Profile</a> |
        <a href="contact.html">Contact</a> |
        <a href="faq.html">FAQ</a>
      </nav>
```

<hr />

</header>

<main>

<section>

<h2>Contact Us</h2>

<p>

If you have any questions, suggestions, or issues, please contact us.

</p>

</section>

<section>

<h3>Contact Information</h3>

<p>Email: support@studenthub.com</p>

<p>Phone: +91 98765 43210</p>

<p>

Address: CHARUSAT University, Changa, Anand,
Gujarat

</p>

</section>

<section>

<h3>Send Your Message</h3>

<form>

<label for="name">Full Name</label>

<input type="text" id="name" name="name" />

<label for="email">Email</label>

<input type="email" id="email" name="email" />

<label for="subject">Subject</label>

<input type="text" id="subject" name="subject" />

<label for="message">Message</label>

<textarea id="message" name="message" rows="5"
cols="40"></textarea>

>

<button type="submit">Send Message</button>

</form>

</section>

</main>

<footer>

<hr />

<p>© 2026 StudentHub | All Rights Reserved.</p>

</footer>

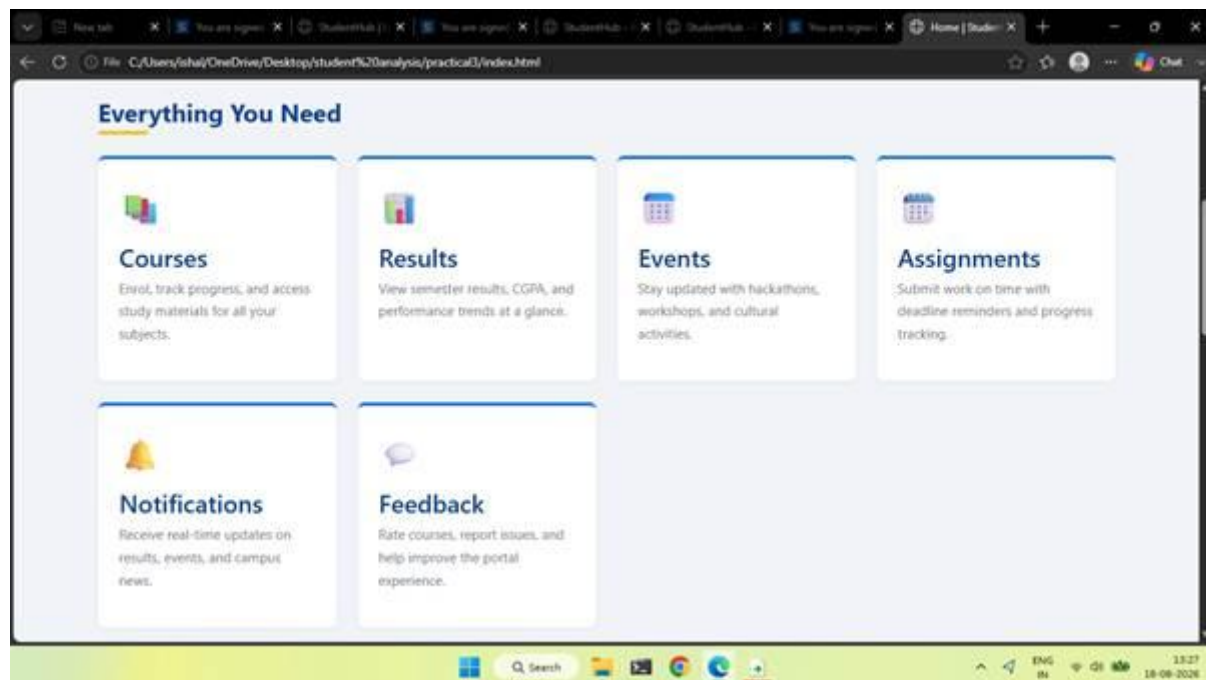
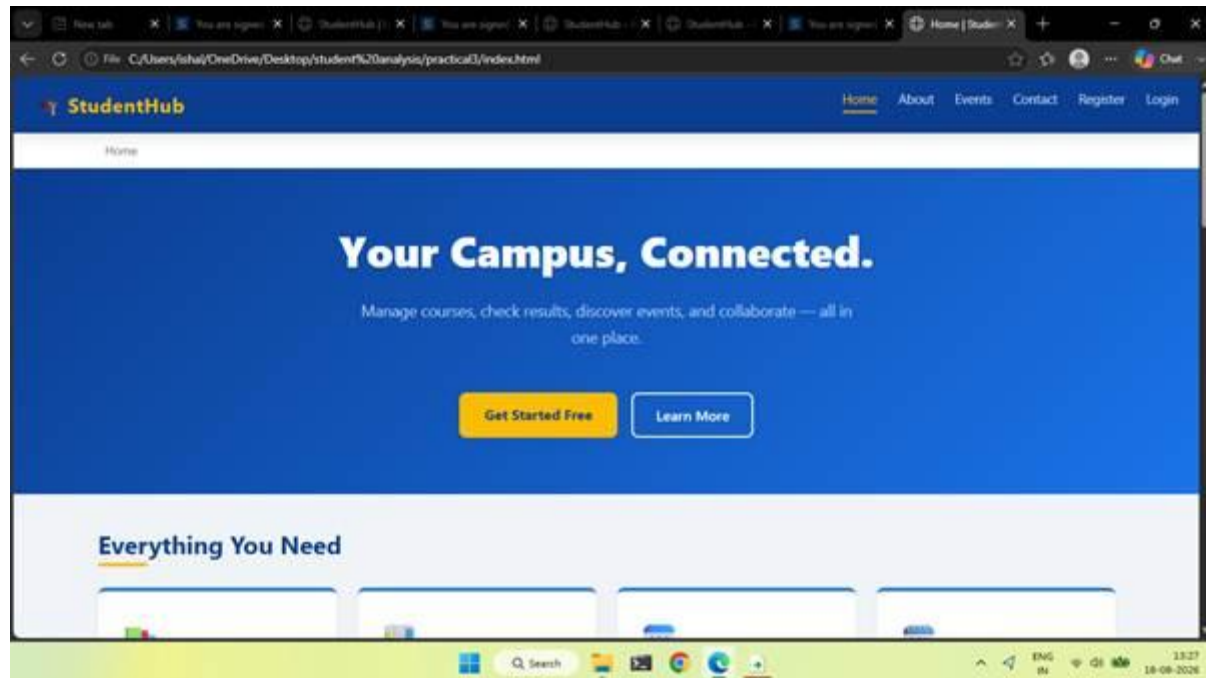
</body>

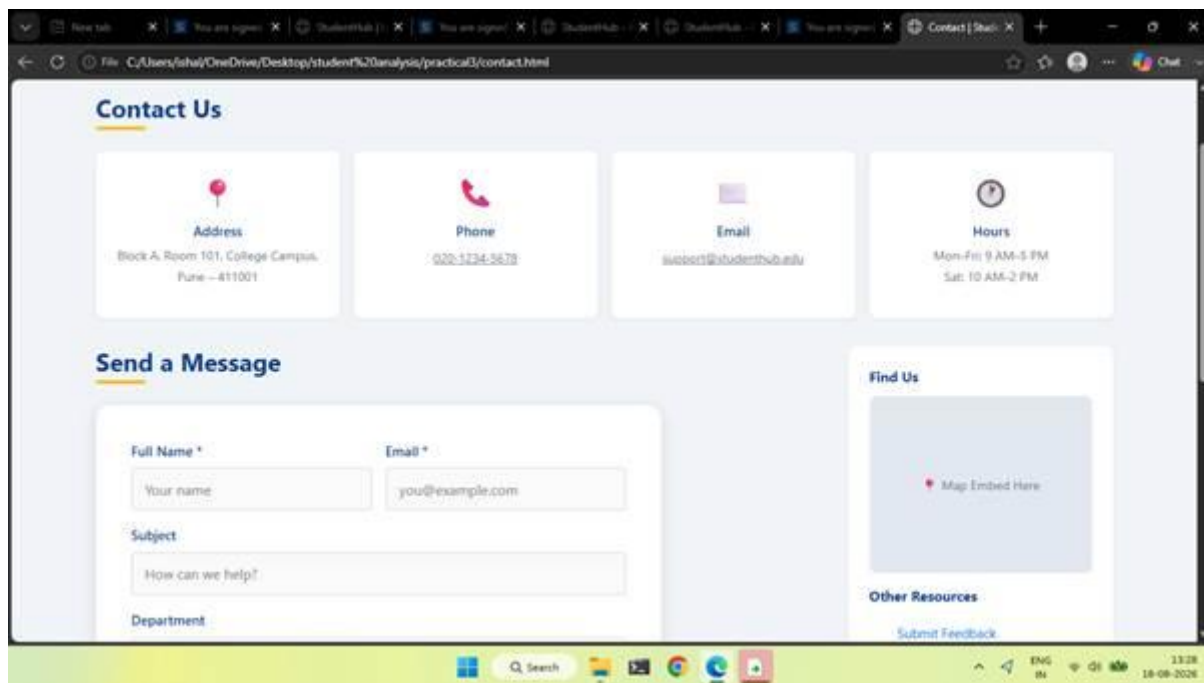
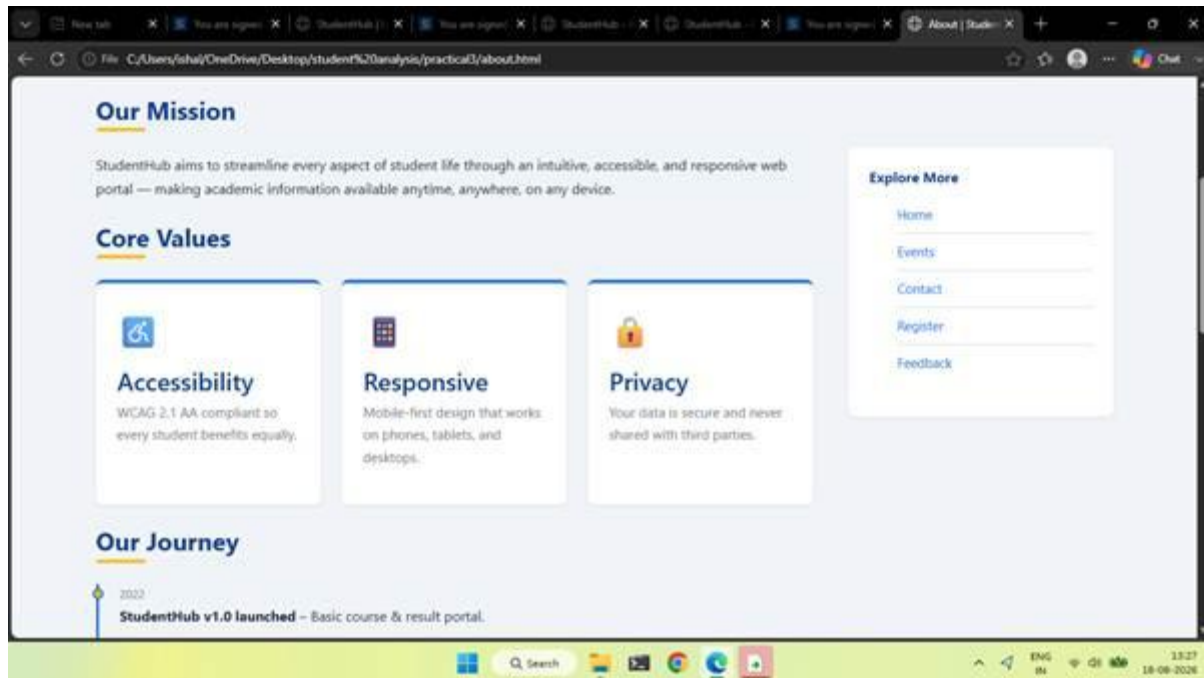
</html>

Output:



Over all combined file output:





Browser tabs: New tab, You are signed out, StudentHub, You are signed out, StudentHub, StudentHub, You are signed out, Register | StudentHub

File: C:\Users\shah\OneDrive\Desktop\student%20analysis\practical3\register.html

Create Your Account

Already registered? [Login here](#)

First Name *

Last Name *

First name

Last name

Email Address *

student@example.com

Mobile Number *

10-digit number

Date of Birth *

dd-mm-yyyy

Course *

-- Select Course --

Year of Study *

-- Select Year --

Gender

-- Select --

Windows taskbar: Search, 13:28, 18-08-2026

Browser tabs: New tab, You are signed out, StudentHub, You are signed out, StudentHub, StudentHub, You are signed out, Login | StudentHub

File: C:\Users\shah\OneDrive\Desktop\student%20analysis\practical3\login.html

StudentHub

Home About Register Login

Welcome Back 🍌

Email Address *

student@example.com

Password *

☐ Remember me [Forgot password?](#)

Login

or continue with

Windows taskbar: Search, 13:28, 18-08-2026